CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circular field has radius 14 m. Fencing costs ₹ 10 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 880 **(b)** ₹ 88 **(c)** ₹ 6160 **(d)** ₹ 440

2. A wheel has radius 21 cm. The number of complete revolutions it makes to cover 66 m is: 1

(a) 25 **(b)** 100 **(c)** 50 **(d)** 60

3. The radii of two circles are 19 cm and 9 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 14 cm **(b)** 28 cm **(c)** 30 cm **(d)** 56 cm

4. The radii of two circles are 6 cm and 8 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 10 cm **(b)** 12 cm **(c)** 2 cm **(d)** 14 cm

5. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 5 minutes is: 1

(a) 115.5 cm² **(b)** 11 cm² **(c)** 231 cm² **(d)** 1386 cm²

6. A horse is tied at a corner of a large square field with a rope 7 m long. The area of the field it can graze is: 1

(a) 77 m² **(b)** 38.5 m² **(c)** 11 m² **(d)** 154 m²

7. The area of a sector of a circle of radius 21 cm is 231 cm^2. The central angle of the sector is: 1

(a) 60° **(b)** 30° **(c)** 90° **(d)** 120°

8. A circular field has radius 21 m. Fencing costs ₹ 5 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 132 **(b)** ₹ 6930 **(c)** ₹ 330 **(d)** ₹ 660

9. A wheel has radius 35 cm. The number of complete revolutions it makes to cover 110 m is: 1

(a) 50 **(b)** 60 **(c)** 100 **(d)** 25

10. The radii of two circles are 8 cm and 6 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 14 cm **(b)** 28 cm **(c)** 10 cm **(d)** 7 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 15 minutes. 2

12. A wire in the form of a circle of radius 14 cm is bent into a square. Find the area of the square. 2

13. A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 14 m. Find the increase in the area it can graze. 2

14. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 30 minutes. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A chord of a circle of radius 7 cm subtends a right angle at the centre. Find the area of the minor segment. 3

16. A sector of a circle of radius 14 cm has central angle 90°. Find the perimeter of the sector. 3

17. An umbrella has 8 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 28 cm. Find the area between two consecutive ribs. 3

OR

A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the minor segment. 3

18. A sector of a circle of radius 14 cm has central angle 180°. Find the perimeter of the sector. 3

OR

An umbrella has 6 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 21 cm. Find the area between two consecutive ribs. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

OR

A brooch is a circular piece of silver wire of diameter 35 mm. The wire is also used for 5 equal diameters that divide the brooch into 10 sectors. Find the total length of wire required (in mm). 5

20. A chord of a circle of radius 21 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

OR

A brooch is a circular piece of silver wire of diameter 28 mm. The wire is also used for 4 equal diameters that divide the brooch into 8 sectors. Find the total length of wire required (in mm). 5

END OF QUESTION PAPER — Total: 40 Marks